SCHOOL OF INFORMATION SCIENCE AND TECHNOLOGY

DEPARTMENT OF LIBRARY SCIENCE

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UNIT NAME: System Analysis and Design

UNIT CODE: BIT 210

CONTINUOUS ASSESSMENT TEST (CAT) ANSWERS

QUESTION 1: System Request (3 Marks)

A system request is a formal proposal submitted to an organisation requesting the development, improvement, replacement or investigation of an information system. It explains why a system-related change is needed and helps management decide whether a project should start.

Two common triggers or key components are:

- A business problem, for example, lost records, delays, errors, high costs or customer complaints.
- A business opportunity or management directive, for example, expanding services, complying with new rules, improving customer service or gaining a competitive advantage.

QUESTION 2: Functional and Non-Functional Requirements (4 Marks)

Functional requirements describe what a system must do. They state the services, processes, business rules, inputs and outputs required by users. Example: “The system shall allow staff to register a parcel and generate a unique tracking number.”

Non-functional requirements describe how well a system must operate or the constraints it must meet. They cover areas such as performance, security, reliability, usability and availability. Example: “The parcel-tracking screen shall return a result within three seconds for at least 95% of requests.”

Therefore, functional requirements focus on system behaviour, while non-functional requirements focus on system quality and operating standards.

QUESTION 3: Joint Application Development, JAD (4 Marks)

Joint Application Development (JAD) is a structured requirements-elicitation technique in which users, managers, systems analysts and technical staff meet in a facilitated workshop.

They discuss the proposed system, identify requirements, resolve disagreements and agree on a common understanding of the system.

Two advantages over traditional one-to-one interviews are:

- Faster requirements gathering: relevant people are in the same meeting, so unclear issues and conflicts can be clarified immediately instead of through repeated follow-up interviews.
- Better consensus and user ownership: participants hear one another's needs, agree on priorities and are more likely to accept the final system because they were involved in defining it.

QUESTION 4: TPS and MIS (4 Marks)

A Transaction Processing System (TPS) mainly serves operational-level staff such as clerks, customer-service officers and dispatchers. Its purpose is to capture, validate, store and process routine day-to-day transactions accurately and quickly. In the CityLink case, parcel registration, payment recording, address updates and delivery-status scans are TPS activities.

A Management Information System (MIS) mainly serves middle-level managers. Its purpose is to transform operational data into organised reports and summaries for monitoring, control and tactical decision-making. For example, a manager can use an MIS report to review late deliveries by route, driver or week.

In summary, TPS records detailed daily transactions, while MIS summarises that information for management decisions.

QUESTION 5: CityLink Express Scenario

The scenario identifies lost package details, delayed dispatches and incorrect delivery addresses caused by semi-manual, handwritten logs. The proposed mobile-enabled tracking system should include the following requirements:

- Digital parcel registration with a unique tracking number.
- Required and validated customer and delivery-address fields to reduce wrong addresses.
- A central database to prevent loss of parcel records.
- Mobile access for drivers to view assignments and update parcel status during collection, transit and delivery.
- Tracking statuses such as registered, dispatched, in transit, out for delivery and delivered.

- User acceptance testing before full rollout to confirm that the system supports the real delivery process.

QUESTION 6: Project Network Paths (3 Marks)

The project starts with Activity A. Once A is completed, Activities B and C can proceed in parallel. Activity D follows B, while Activity E can start only after both C and D are completed.

The possible start-to-finish paths are:

- Path 1: A → B → D → E
- Path 2: A → C → E

QUESTION 7: Critical Path and Minimum Duration (4 Marks)

Path A → B → D → E takes: 2 + 3 + 4 + 2 = 11 weeks.

Path A → C → E takes: 2 + 2 + 2 = 6 weeks.

The critical path is A → B → D → E because it has the longest duration. Therefore, the minimum project completion time is 11 weeks. Any delay in Activities A, B, D or E delays the whole project. Activity C has 5 weeks of float, provided it is completed before Activity E begins.

QUESTION 8: Chen ER Model (4 Marks)

In Chen notation, CUSTOMER and PARCEL are represented as rectangles because they are entities. They are linked by a diamond labelled PLACES, or SENDS, because the diamond represents the relationship.

The cardinality is one-to-many (1:M): one CUSTOMER can place many PARCELS, but each PARCEL belongs to one CUSTOMER. Write 1 next to CUSTOMER and M or N next to PARCEL.

Attributes are represented by ovals connected to their entity. CUSTOMER can have Customer_ID, Name, Phone_Number and Address. PARCEL can have Parcel_ID, Tracking_Number, Destination, Weight and Status. Key attributes such as Customer_ID and Parcel_ID are underlined. In a relational database, Customer_ID would normally appear in the PARCEL table as a foreign key.

QUESTION 9: Direct and Parallel Changeover (4 Marks)

Direct changeover means the old semi-manual system is stopped and the new automated system begins immediately. It is fast and relatively cheap because only one system is operating. Its key risk is high operational disruption: if the new system fails or staff are not ready, CityLink Express may have no working fallback and delivery operations may stop.

Parallel changeover means the old and new systems run together for a temporary period. The organisation compares results and retires the old system only when the new one is proven reliable. It is safer because the old method remains available as a fallback. Its key risk is additional cost and workload, since staff may need to enter the same parcel data twice and inconsistent records may arise if the two systems are not updated in the same way.